



Given: Due:	Further Maths (Core Pure) HW 24	Name:
Chapter 9: Vector lines, intersections & angles		
Q1. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. The line l_2 has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$. (a) Show that l_1 and l_2 do not meet. (4) The point A is on l_1 where $\lambda = 1$, and the point B is on l_2 where $\mu = 2$. (b) Find the cosine of the acute angle between AB and l_1. (6)		

In part (a), a majority of candidates were able to prove that the two lines did not cross, although some candidates produced errors in solving relatively straightforward "simultaneous" equations. A small number of candidates tried to show a contradiction by substituting their values for λ and μ into one of the two equations they had already used, sometimes with apparent success, as they had already found an incorrect value of one of the parameters! There were a few candidates, however, who believed that the two lines were parallel and attempted to prove this.

Part (b) was less well answered. Many candidates found $\overrightarrow{OA}$ and $\overrightarrow{OB}$, although there was a surprising number of numeric errors seen in finding these vectors. A significant proportion of candidates did not subtract these vectors in order to find $\overrightarrow{AB}$. Most candidates then knew that a dot product was required but there was great confusion on which two vectors to use. A significant minority correctly applied the dot product formula between $\overrightarrow{AB}$ and the direction vector of l_1 . Common errors here included applying the dot product formula between either $\overrightarrow{OA}$ and $\overrightarrow{OB}$; or the direction vector of l_1 and the direction vector of l_2 ; or the direction vector of l_1 and twice the direction vector of l_2 ; or $\overrightarrow{AB}$ and the direction vector of l_2 . A few candidates did not state the cosine of the acute angle as question required but instead found the acute angle.



Question Number	Scheme	Marks
(a)	If l_1 and l_2 intersect then: $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ i: $1 + \lambda = 1 + 2\mu$ (1) Any two of j: $\lambda = 3 + \mu$ (2) k: $-1 = 6 - \mu$ (3) (1) & (2) yields $\lambda = 6, \mu = 3$ (1) & (3) yields $\lambda = 14, \mu = 7$ (2) & (3) yields $\lambda = 10, \mu = 7$ checking eqn (3), $-1 \neq 3$ Either checking eqn (2), $14 \neq 10$ checking eqn (1), $11 \neq 15$ or for example: checking eqn (3), LHS = -1, RHS = 3 $\Rightarrow$ Lines l_1 and l_2 do not intersect or for example: checking eqn (3), LHS = -1, RHS = 3 $\Rightarrow$ Lines l_1 and l_2 do not intersect	Writes down any two of these equations correctly. M1 Solves two of the above equations to find ... either one of λ or μ correct A1 both λ and μ correct A1 Complete method of putting their values of λ and μ into a third equation to show a contradiction. B1 $\sqrt{}$ this type of explanation is also allowed for B1 $\sqrt{}$. [4]
Aliter (a) Way 2	k: $-1 = 6 - \mu \Rightarrow \mu = 7$ i: $1 + \lambda = 1 + 2\mu \Rightarrow 1 + \lambda = 1 + 2(7)$ j: $\lambda = 3 + \mu \Rightarrow \lambda = 3 + (7)$ i: $\lambda = 14$ j: $\lambda = 10$ Either: These equations are then inconsistent Or: $14 \neq 10$ Or: Lines l_1 and l_2 do not intersect	Uses the k component to find μ and substitutes their value of μ into either one of the i or j component. M1 either one of the λ's correct A1 both of the λ's correct A1 Complete method giving rise to any one of these three explanations. B1 $\sqrt{}$ [4]
Aliter (a) Way 3	If l_1 and l_2 intersect then: $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ i: $1 + \lambda = 1 + 2\mu$ (1) Any two of j: $\lambda = 3 + \mu$ (2) k: $-1 = 6 - \mu$ (3) (1) & (2) yields $\mu = 3$ (3) yields $\mu = 7$ Either: These equations are then inconsistent Or: $3 \neq 7$ Or: Lines l_1 and l_2 do not intersect	Writes down any two of these equations M1 either one of the μ's correct A1 both of the μ's correct A1 Complete method giving rise to any one of these three explanations. B1 $\sqrt{}$ [4]
Aliter (a) Way 4	i: $1 + \lambda = 1 + 2\mu$ (1) Any two of j: $\lambda = 3 + \mu$ (2) k: $-1 = 6 - \mu$ (3) (1) & (2) yields $\mu = 3$ (3) RHS = $6 - 3 = 3$ (3) yields $-1 \neq 3$	Writes down any two of these equations M1 $\mu = 3$ A1 RHS of (3) = 3 A1 Complete method giving rise to this explanation. B1 $\sqrt{}$ [4]



(b)	$\lambda = 1 \Rightarrow \overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \quad \& \quad \mu = 2 \Rightarrow \overrightarrow{OB} = \begin{pmatrix} 5 \\ 5 \\ 4 \end{pmatrix}$	Only one of either $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \text{ or } \overrightarrow{OB} = \begin{pmatrix} 5 \\ 5 \\ 4 \end{pmatrix} \text{ or}$ $A(2, 1, -1) \text{ or } B(5, 5, 4).$ (can be implied)	B1 M1√ M1 M1√ A1 A1 cao [6] 10 marks
	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5 \\ 5 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \text{ or } \overrightarrow{BA} = \begin{pmatrix} -3 \\ -4 \\ -5 \end{pmatrix}$	Finding the difference between their $\overrightarrow{OB}$ and $\overrightarrow{OA}$. (can be implied)	
	$\overrightarrow{AB} = 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}, \mathbf{d}_1 = \mathbf{i} + \mathbf{j} + 0\mathbf{k} \text{ \& \& } \theta \text{ is angle}$	Applying the dot product formula between "allowable" vectors. See notes below.	
	$\cos \theta = \frac{\overrightarrow{AB} \cdot \mathbf{d}_1}{ \overrightarrow{AB} \mathbf{d}_1 } = \pm \left(\frac{3 + 4 + 0}{\sqrt{50} \cdot \sqrt{2}} \right)$	Applies dot product formula between $\mathbf{d}_1$ and their $\pm \overrightarrow{AB}$. Correct expression.	
	$\cos \theta = \frac{7}{10}$	$\frac{7}{10}$ or 0.7 or $\frac{7}{\sqrt{100}}$ but not $\frac{7}{\sqrt{50} \cdot \sqrt{2}}$	

Candidates can score this mark if there is a complete method for finding the dot product between their vectors in the following cases:

Case 1: their ft $\pm \overrightarrow{AB} = \pm(3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k})$
and $\mathbf{d}_1 = \mathbf{i} + \mathbf{j} + 0\mathbf{k}$
 $\Rightarrow \cos \theta = \pm \left(\frac{3 + 4 + 0}{\sqrt{50} \cdot \sqrt{2}} \right)$

Case 2: $\mathbf{d}_1 = \mathbf{i} + \mathbf{j} + 0\mathbf{k}$
and $\mathbf{d}_2 = 2\mathbf{i} + \mathbf{j} - 1\mathbf{k}$
 $\Rightarrow \cos \theta = \frac{2 + 1 + 0}{\sqrt{2} \cdot \sqrt{6}}$

Case 3: $\mathbf{d}_1 = \mathbf{i} + \mathbf{j} + 0\mathbf{k}$
and $\mathbf{d}_2 = 2(2\mathbf{i} + \mathbf{j} - 1\mathbf{k})$
 $\Rightarrow \cos \theta = \frac{4 + 2 + 0}{\sqrt{2} \cdot \sqrt{24}}$

Case 4: their ft $\pm \overrightarrow{AB} = \pm(3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k})$
and $\mathbf{d}_2 = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$
 $\Rightarrow \cos \theta = \pm \left(\frac{6 + 4 - 5}{\sqrt{50} \cdot \sqrt{6}} \right)$

Case 5: their ft $\overrightarrow{OA} = 2\mathbf{i} + 1\mathbf{j} - 1\mathbf{k}$
and their ft $\overrightarrow{OB} = 5\mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$
 $\Rightarrow \cos \theta = \pm \left(\frac{10 + 5 - 4}{\sqrt{6} \cdot \sqrt{66}} \right)$

Note: If candidate use cases 2, 3, 4 and 5 they cannot gain the final three marks for this part.
Note: Candidate can only gain some/all of the final three marks if they use case 1.

Examples of awarding of marks M1M1A1 in 5.(b)	
Example	Marks
$\sqrt{50} \cdot \sqrt{2} \cos \theta = \pm(3 + 4 + 0)$	M1M1A1 (Case 1)
$\sqrt{2} \cdot \sqrt{6} \cos \theta = 3$	M1M0A0 (Case 2)
$\sqrt{2} \cdot \sqrt{24} \cos \theta = 4 + 2$	M1M0A0 (Case 3)

Q2.

Relative to a fixed origin O , the point A has position vector $(10\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$, and the point B has position vector $(8\mathbf{i} + 3\mathbf{j} + 4\mathbf{k})$.

The line l passes through the points A and B .

(a) Find the vector $\overrightarrow{AB}$.

(2)

(b) Find a vector equation for the line l .

(2)

The point C has position vector $(3\mathbf{i} + 12\mathbf{j} + 3\mathbf{k})$.

The point P lies on l . Given that the vector $\overrightarrow{CP}$ is perpendicular to l ,

(c) find the position vector of the point P .

(6)

In part (a), both marks were gained by nearly all candidates. Only a minority added the components or subtracted them the wrong way round. If $\overrightarrow{BA}$ was given instead of $\overrightarrow{AB}$, one mark was lost here but all other marks in the question could be gained. Again, in part (b), most candidates gained both marks but a substantial number of candidates failed to realise that the equation of a line must actually be an equation and,

for example, omitted the "r =" from $\mathbf{r} = \begin{pmatrix} 8 \\ 3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$.

Part (c) was very demanding and often proved to be an all or nothing affair. Correct methods were not often seen but, when they were, full marks were usually gained. Most knew that they should equate a scalar product to zero but the wrong vectors were often used. It would have helped candidates if they had drawn a diagram showing the points O , A , B , C and P . The scalar products $\overrightarrow{OP} \cdot \overrightarrow{AB}$ or $\overrightarrow{OP} \cdot \overrightarrow{OC}$ were often used instead of $\overrightarrow{CP} \cdot \overrightarrow{AB}$. To make progress with this question it is necessary to use the correct scalar product to obtain an equation in a single parameter, not the three unknowns x , y and z . The most commonly seen successful method was to use the candidate's answer to (b) to obtain a vector expression for $\overrightarrow{CP}$ in terms of, say, λ . If the

$$\overrightarrow{CP} = \begin{pmatrix} 5 - 2\lambda \\ \lambda - 9 \\ \lambda + 1 \end{pmatrix}$$

equation of the line is that given in the previous paragraph, this gives $\overrightarrow{CP} \cdot \overrightarrow{AB} = 0$. The scalar product of this vector with the answer to part (a) gives an equation for λ . The value of the parameter given by this equation can then be substituted into the answer to part (b) to complete the question. There are alternative approaches to this question using differentiation or using the cosine rule to find the ratio, for example, $AB : AP$ but these were very rarely seen.



Question Number	Scheme	Marks
	(a) $\vec{AB} = \begin{pmatrix} 8 \\ 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 10 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ (b) $\mathbf{r} = \begin{pmatrix} 10 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ $\mathbf{r} = \begin{pmatrix} 8 \\ 3 \\ 4 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ (c) $\vec{CP} = \begin{pmatrix} 10-2t \\ 2+t \\ 3+t \end{pmatrix} - \begin{pmatrix} 3 \\ 12 \\ 3 \end{pmatrix} = \begin{pmatrix} 7-2t \\ t-10 \\ t \end{pmatrix}$ $\begin{pmatrix} 7-2t \\ t-10 \\ t \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = -14 + 4t + t - 10 + t = 0$ Leading to $t = 4$ Position vector of P is $\begin{pmatrix} 10-8 \\ 2+4 \\ 3+4 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 7 \end{pmatrix}$ Alternative working for (c) $\vec{CP} = \begin{pmatrix} 8-2t \\ 3+t \\ 4+t \end{pmatrix} - \begin{pmatrix} 3 \\ 12 \\ 3 \end{pmatrix} = \begin{pmatrix} 5-2t \\ t-9 \\ t+1 \end{pmatrix}$ $\begin{pmatrix} 5-2t \\ t-9 \\ t+1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = -10 + 4t + t - 9 + t + 1 = 0$ Leading to $t = 3$ Position vector of P is $\begin{pmatrix} 8-6 \\ 3+3 \\ 4+3 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 7 \end{pmatrix}$	M1 A1 (2) M1 A1ft (2) M1 A1 M1 A1 M1 A1 (6) [10] M1 A1 M1 A1 M1 A1 (6)

Q3.

Relative to a fixed origin O , the point A has position vector $(8\mathbf{i} + 13\mathbf{j} - 2\mathbf{k})$,
 the point B has position vector $(10\mathbf{i} + 14\mathbf{j} - 4\mathbf{k})$,
 and the point C has position vector $(9\mathbf{i} + 9\mathbf{j} + 6\mathbf{k})$.

The line l passes through the points A and B .

(a) Find a vector equation for the line l .

(3)

(b) Find $|\vec{CB}|$.

(2)

(c) Find the size of the acute angle between the line segment CB and the line l , giving your answer in degrees to 1 decimal place.

(3)

Part (a) was quite well answered, most finding $\vec{AB}$ or $\vec{BA}$

and writing down $\vec{OA} + \lambda \vec{AB}$, or an equivalent. An equation does, however need an equals sign and a subject and many lost the final A mark in this part by omitting the "r =" from, say, $\mathbf{r} = 8\mathbf{i} + 13\mathbf{j} - 2\mathbf{k} + \lambda(2\mathbf{i} + \mathbf{j} - 2\mathbf{k})$. In part (b), those who realised that a magnitude or length was required were usually successful. In part (c), nearly all candidates knew how to evaluate a scalar product and obtain an equation in $\cos\theta$, and so gain the method marks, but the vectors chosen were not always the right ones and a few candidates gave the obtuse angle.

Question Number	Scheme	Marks
Q (a)	$\vec{AB} = \vec{OB} - \vec{OA} = \begin{pmatrix} 10 \\ 14 \\ -4 \end{pmatrix} - \begin{pmatrix} 8 \\ 13 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ $\text{or } \vec{BA} = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$ $\mathbf{r} = \begin{pmatrix} 8 \\ 13 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 10 \\ 14 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ accept equivalents	M1 M1 A1ft (3)
(b)	$\vec{CB} = \vec{OB} - \vec{OC} = \begin{pmatrix} 10 \\ 14 \\ -4 \end{pmatrix} - \begin{pmatrix} 9 \\ 9 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ -10 \end{pmatrix}$ $\text{or } \vec{BC} = \begin{pmatrix} -1 \\ -5 \\ 10 \end{pmatrix}$ $CB = \sqrt{1^2 + 5^2 + (-10)^2} = \sqrt{126} \quad (= 3\sqrt{14} \approx 11.2)$ awrt 11.2	M1 A1 (2)
(c)	$\vec{CB} \cdot \vec{AB} = \vec{CB} \vec{AB} \cos \theta$ $(\pm)(2 + 5 + 20) = \sqrt{126} \sqrt{9} \cos \theta$ $\cos \theta = \frac{3}{\sqrt{14}} \Rightarrow \theta \approx 36.7^\circ$ awrt 36.7°	M1 A1 A1 (3)



Q4.

$$\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix},$$

The line l_1 has equation where λ is a scalar parameter.

$$\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix},$$

The line l_2 has equation where μ is a scalar parameter.

Given that l_1 and l_2 meet at the point C , find

(a) the coordinates of C .

(3)

The point A is the point on l_1 where $\lambda = 0$ and the point B is the point on l_2 where $\mu = -1$.

(b) Find the size of the angle ACB . Give your answer in degrees to 2 decimal places.

(4)

(c) Hence, or otherwise, find the area of the triangle ABC .

(5)

Part (a) was fully correct in the great majority of cases but the solutions were often unnecessarily long and nearly two pages of working were not unusual. The simplest method is to equate the $\mathbf{j}$ components. This gives one equation in λ , leading to $\lambda = 3$, which can be substituted into the equation of l_1 to give the coordinates of C . In practice, the majority of candidates found both λ and μ and many proved that the lines were coincident at C . However the question gave the information that the lines meet at C and candidates had not been asked to prove this. This appeared to be another case where candidates answered the question that they had expected to be set, rather than the one that actually had been.

The great majority of candidates demonstrated, in part (b), that they knew how to find the angle between two vectors using a scalar product. However the use of the position vectors of A and B , instead of vectors in the directions of the lines was common. Candidates could have used either

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} 5 \\ 0 \\ 2 \end{pmatrix},$$

the vectors given in the question, or $\overrightarrow{AC}$ and $\overrightarrow{BC}$. The latter was much the commoner choice but many made errors in signs. Comparatively few chose to use the cosine rule. In part (c), many continued with the position vectors they had used incorrectly in part (b) and so found the area of the triangle OAB rather than triangle ABC . The easiest method of completing part (c) was usually to use the formula $\text{Area} = \frac{1}{2}ab \sin C$ and most chose this.

Attempts to use $\text{Area} = \frac{1}{2} \text{base} \times \text{height}$ were usually fallacious and often assumed that the triangle was isosceles. A few complicated attempts were seen which used vectors to find the coordinates of the foot of a perpendicular from a vertex to the opposite side. In principle, this is possible but, in this case, the calculations proved too difficult to carry out correctly under examination conditions.



Question Number	Scheme	Marks
	(a) j components $3 + 2\lambda = 9 \Rightarrow \lambda = 3$ ($\mu = 1$) Leading to $C : (5, 9, -1)$ accept vector forms (b) Choosing correct directions or finding $\overline{AC}$ and $\overline{BC}$ $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 0 \\ 2 \end{pmatrix} = 5 + 2 = \sqrt{6} \sqrt{29} \cos \angle ACB$ use of scalar product $\angle ACB = 57.95^\circ$ awrt 57.95° (c) $A : (2, 3, -4)$ $B : (-5, 9, -5)$ $\overline{AC} = \begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix}$, $\overline{BC} = \begin{pmatrix} 10 \\ 0 \\ 4 \end{pmatrix}$ $AC^2 = 3^2 + 6^2 + 3^2 \Rightarrow AC = 3\sqrt{6}$ $BC^2 = 10^2 + 4^2 \Rightarrow BC = 2\sqrt{29}$ $\Delta ABC = \frac{1}{2} AC \times BC \sin \angle ACB$ $= \frac{1}{2} 3\sqrt{6} \times 2\sqrt{29} \sin \angle ACB \approx 33.5$ $15\sqrt{5}$, awrt 34	M1 A1 A1 (3) M1 M1 A1 A1 (4) M1 A1 A1 M1 A1 (5) [12]
	<i>Alternative method for (b) and (c)</i> (b) $A : (2, 3, -4)$ $B : (-5, 9, -5)$ $C : (5, 9, -1)$ $AB^2 = 7^2 + 6^2 + 1^2 = 86$ $AC^2 = 3^2 + 6^2 + 3^2 = 54$ $BC^2 = 10^2 + 0^2 + 4^2 = 116$ Finding all three sides $\cos \angle ACB = \frac{116 + 54 - 86}{2\sqrt{116}\sqrt{54}} (= 0.53066 \dots)$ $\angle ACB = 57.95^\circ$ awrt 57.95° If this method is used some of the working may gain credit in part (c) and appropriate marks may be awarded if there is an attempt at part (c).	M1 M1 A1 A1 (4)